

End Of Year Project EYP fo the student Amira Jemaa

1. Project Title and Description

Title: Local Voice AI Dev Assistant with Project Context.

Description: Development of a desktop application capable of listening to a developer's voice instructions (e.g., "Explain what this function does" or "Generate a unit test for this file") using AI models running locally on the machine to ensure code confidentiality.

2. Problem Statement and Motivations

Problem Statement

While developers increasingly use AI tools like GitHub Copilot, two major issues remain:

1. **Privacy:** Sending proprietary code to third-party servers is often prohibited in corporate environments.
2. **Workflow Accessibility:** Constant text-based interaction can break the creative "flow." How can we provide hands-free, secure assistance?

Motivations

- **Total Confidentiality:** Implementing "On-Device" (Edge AI) models.
- **Productivity:** Allowing the developer to ask questions out loud while staying focused on their code editor.
- **Skill Acquisition:** Mastering the integration of LLMs (Large Language Models) and Speech Processing (STT/TTS).

3. Core Architecture and Features

The student should focus on three manageable modules:

1. **Transcription Module (STT):** Converting the developer's speech into text.
2. **Intelligence Module (LLM):** Processing the text by injecting the content of the open code file into the prompt.
3. **Interface Module:** A simple floating window or a basic extension to display the response.

4. Tech Stack (Suggested Tools)

The choice of tools focuses on high-level libraries with strong community support to ensure the student can find documentation easily.

Component	Recommended Tool	Why?
Language	Python	Essential for AI and easy to learn for GLSI students.
Interface	CustomTkinter or Streamlit	Fast and modern GUI creation using pure Python.
AI Engine (Local)	Ollama	Allows running models like <i>Llama 3</i> or <i>Mistral</i> with a single command.
Transcription	OpenAI Whisper (Local)	The most robust model for local speech recognition, available for free.
Context Handling	LangChain	Simplifies reading project files and feeding them to the model.

5. Implementation Roadmap

- **Phase 1:** Setting up Ollama and testing text queries on code snippets (Basic RAG).
- **Phase 2:** Integrating Whisper to transform voice input into text.
- **Phase 3:** Developing the GUI that allows the user to select the project directory.
- **Phase 4:** Unit testing and thesis writing.